

WeVo Pay

P2P Financial Exchange Platform

Final Project Documentation

Egypt Digital Pioneers Initiative

Team Members

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1. Project Idea

Many people in Egypt need to move money between mobile wallets like Vodafone Cash or Orange Money and bank accounts through InstaPay. Right now, there is no easy and safe official way to do this. Most people end up asking strangers in Facebook or Telegram groups to help them with the transfer. This is risky because a lot of people get scammed this way.

WeVo Pay was created to solve this problem. It is a platform where users can post offers to exchange money between wallets and InstaPay. When someone accepts an offer, both users go into a private Trade Room. There, they send the money to each other directly and upload screenshots as proof. Only after both sides confirm that they received the money does the system mark the transaction as complete. This way, the platform acts like a safe middleman without actually holding the money itself.

The platform also has a small fee of 0.5% that is taken only when the transaction finishes successfully. Users can also earn free transactions after completing a certain number of trades.

2. What We Are Building

We are building a web application where users can safely do wallet-to-InstaPay exchanges. The main parts of the system are:

Registration and Login

Users start by creating an account using their email. They verify their email with a code sent to them (OTP). After that, they complete their profile by adding their username, birth date, governorate, gender, address, and phone number. They can also add a referral code if someone invited them. This information is stored in the database so we know who is using the platform.

KYC Verification

Before users can create or accept offers, they must verify their identity. This is called KYC. They upload a photo of their National ID and also take a live selfie using their camera. This helps make sure the person using the account is real and reduces the chance of fake accounts or fraud.

Market Feed and Offers

Verified users can create offers. For example, someone who has money in Vodafone Cash and wants to receive it in InstaPay can post an offer with the amount and the commission they are willing to pay. Other users can browse these offers in the market feed. If they see an offer that matches what they need (for example, they have InstaPay and want cash in a wallet), they can accept it.

Trade Room

When an offer is accepted, it gets locked so no one else can take it. Both users then enter a private Trade Room made just for them. In this room, they talk to each other if needed, send the money directly to each other's wallet or bank account, and upload screenshots of the successful transfer as proof. After both users upload their receipts and confirm, the transaction is marked as completed.

Profile and Earnings

Users can see their profile where they track how much money they earned from transactions, how many trades they did, and whether they qualified for free transactions. There is also a referral system where users can invite others and get benefits when those people complete trades.

Disputes

If something goes wrong during a transaction (for example, one side claims they sent the money but the other side says they did not receive it), any user can raise a dispute. This sends the case to the admin team who can review the receipts and messages and decide what to do.

3. System Design

The application is built using ASP.NET Core MVC. This means we have Models that represent the data in the database, Controllers that handle the logic, and Views that show the pages to the user. We used Entity Framework Core to connect to SQL Server and manage the database.

Main Tables in the Database

We have a User table that stores all user information like name, email, phone, password (stored securely), governorate, gender, KYC photo path, and whether the account is verified. The Offer table holds the offers users create, including the amount, what wallet they are using, what they want to receive, and the current status of the offer. When two users start trading, we create a record in the Transaction table. This table keeps track of the two users involved, the original offer, and the paths to the receipt images they uploaded. If there is a problem, a Dispute record is created that links to the transaction and records what the user complained about.

How the Main Flow Works

A user registers and verifies their email. Then they complete KYC. After that they can create an offer or browse offers created by others. When they accept an offer, the system locks it and creates a transaction between them and the offer creator. Both users go to the Trade Room page where they can upload their receipt images. Once both sides confirm they received the money, the transaction is finished and the fee is recorded. If there is any issue, they can open a dispute instead.

4. Implementation

On the backend, we created models for User, Offer, Transaction, and Dispute. We set up the database context and used migrations to create the tables. Controllers were written to handle registration, login, KYC upload, offer creation, accepting offers, managing the Trade Room, and handling disputes. File uploads were done using IFormFile, and images are saved on the server with unique names.

On the frontend side, we used Razor Views with Bootstrap to build the pages. The registration has multiple steps. The KYC page supports both file upload and live camera capture. The market feed shows offers in cards, and the Trade Room shows the two users, receipt upload areas, and confirmation buttons. Some JavaScript was used to make the forms and buttons work better.

5. Team

Four people worked on this project. Two people focused on the backend development, which includes the database, controllers, and core logic. The other two people worked on the frontend, building the user interface screens and making sure the pages look good and work smoothly for the user.

6. Testing

We tested the main features step by step. This included creating accounts, verifying with OTP, completing KYC, creating offers, accepting offers, uploading receipts in the Trade Room, and completing transactions. We also tested what happens when someone tries to accept an offer that was already taken, and made sure only the correct users can enter a Trade Room. Problems we found were fixed before the final version.